

GeNERscore: Semantic Alignment Metrics for Generative Named Entity Recognition

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Abstract

Traditional Named Entity Recognition (NER) metrics rely on syntactic span and type matching. While this may be appropriate if NER is framed as a sequence labeling task, the advent of generative models allows for alternative approaches, where model outputs are not extracted, but generated. In this perspective, generative NER calls for more semantically-aware scoring mechanisms. Generative models may produce valid aliases, paraphrases, or type variants that differ from the annotated gold, while also producing unsupported predictions. We propose a taxonomy of such cases and introduce an embedding-based parametric method to evaluate generative NER based on three metrics: *soft precision*, *soft recall* and *unsupported rate*. We validate this framework using standard and noisy datasets, across state-of-the-art systems such as GPT and GLiNER alongside older encoder-based models. We find that our metrics mostly agree with traditional ones on better-established, less noisy benchmarks, but differ in more recent, noisier ones, and can capture edge cases that arise from free-form generation.

1 Introduction

Named Entity Recognition (NER) aims at recognizing entities, each described by a *mention*, i.e., a surface-form span of text, and a *type*, i.e., the semantic class assigned to it. NER is typically evaluated by comparing the entity spans extracted by a model against gold-standard annotations. Traditionally, a prediction is correct only when both its boundary and type match the gold annotation exactly, a convention that runs from the MUC evaluations through CoNLL-2003 (Chinchor, 1995; Tjong Kim Sang and De Meulder, 2003). To relax the rigidity of exact-matching, SemEval-2013 introduced softer views of correctness, including partial and type matching, which separate boundary errors from type errors (Segura-Bedmar et al.,

2013). These views recognize near misses, but they remain surface-bound: they decide correctness through span identity or overlap rather than semantic equivalence. Accordingly, they cannot credit aliases, paraphrases, or descriptive mentions that do not overlap the gold span, and may over-credit predictions with a high overlap with the gold span but no semantic correspondence with it.

Generative models introduce further complications. Unlike sequence taggers, generative models are not restricted to copying input spans. Instead, they may normalize or paraphrase a mention, or choose a different label name. These could be useful, welcome features, but they harm the existing evaluation methods. Consider a model that predicts *JFK* for the gold entity *John F. Kennedy*. They name the same person, but share no contiguous substring, so any kind of syntactic matching, exact or partial, would score the prediction as wrong, which may be too penalizing. To complicate things further, generative models are also prone to hallucinations, therefore the generated text may be unrelated to the source. String matching cannot distinguish between a semantically sound rephrase and hallucinated text. These mismatches are most visible on noisy, user-generated text, where entities typically appear under many aliases or with rare and unstable surface forms (Baldwin et al., 2015; Derczynski et al., 2017). We posit that noisy text should not be treated as a special case, but as the general form NER inputs may take, and that NER evaluation should reflect this setting.

We propose Generative NER score (**GeNERscore**), an embedding-based and entity-level view of NER evaluation, inspired by semantic metrics for text generation that complement exact overlap with contextual similarity (Zhang et al., 2020). Rather than matching strings, **GeNERscore** represents gold and predicted typed entities in the same embedding space, matches them with a configurable rule, and

scores such an alignment with soft analogues of precision and recall. Alongside these metrics, we also propose an indicator of over-generated entities.

We make three major contributions:

- We articulate a taxonomy of evaluation mismatches for generative NER.
- We introduce **GeNERscore**, a semantic alignment framework that compares predicted and gold entities through embedding similarity and configurable assignment algorithms.
- We evaluate **GeNERscore** on four datasets and three model families, showing that it agrees with exact and partial F1 on discriminative models and recovers credit withheld by string matching on generative ones.

2 Motivation

To motivate our contribution, we illustrate six non-mutually exclusive categories of phenomena where string-based evaluation fails to account for meaning equivalence. For each category, we report one example, taken from predictions that strict matching scored as wrong across our datasets and models (Section 5) but could be acceptable, semantically speaking. These examples are illustrative and are not claimed to be statistically representative; they are chosen to show generative models’ behaviour and how current evaluation can be overly penalizing for them.

*“For Taiwan firms in Dongguan, investing in mainland **China** is a road of no return.”*
— OntoNotes 5 #6454

Alias and surface-form variation. The gold annotates China/GPE, while both GPT 5.4-mini and GLiNER2 predict mainland China/GPE. Exact matching scores this as wrong and partial overlap recovers some credit, yet the prediction names the same place. A semantically-aware metric should reward it.

*“24 days until @Cyberpunk 2077@** comes out. I am ready baby !”*
— TweetNER7 #476

Type disagreement. The span matches exactly and only the type label differs. The gold annotates @Cyberpunk 2077@/product, while GPT 5.4-mini and GLiNER models predict Cyberpunk 2077/creative_work, two categories that overlap

heavily, as in this span, which qualifies as both. Partial F1 collapses to zero on this span, even though the labels are similar. A semantically-aware metric should take into account type similarity too, without over-penalizing near-correct predictions.

*“Give a Candlestick **Park card for the memories!** #SFGiants {{URL}} via {@Etsy@}”*
— TweetNER7 #2245

Semantic paraphrase. The gold annotates Park card for the memories/product, while GLiNER model predicts Candlestick Park card/product. The two annotations name the same item, but the surface forms differ enough that exact match and most partial matching methods score it as wrong despite the equivalent meaning. A semantically-aware metric should account for this equivalence.

*“Result of an **English F.A. Challenge**”*
— CoNLL-2003 #139

Partial-overlap over-credit. The gold marks English F.A. Challenge/MISC while the encoder-based model predicts English/MISC, dropping most of the name. While they differ in meaning, a partial match would reward this prediction. A semantically-aware metric should credit the prediction proportionally to its semantic equivalence to the gold annotation. In this case, penalize entirely as they differ.

“... WE LOVE YOU SOHEL #HBDIs-martSohel #HBDSyedSohelRyan”
— TweetNER7 #2711

Annotation incompleteness. The gold annotates only the two hashtags as person mentions, whereas GPT 5.4-mini model also predicts SOHEL/person, a named individual the gold omits. The unannotated entity counts as a false positive against string-based metrics and penalizes precision.

“Jill Biden Leaves the White House in Green Jewel-Toned Jacket & Her Favorite Boot Trend {{URL}} via {@Yahoo@} Hey {@MELANIA TRUMP@}, did you see what Dr. Biden was wearing the other day ?”
— TweetNER7 #2414

Hallucination. Among several correct predictions, GPT 5.4-mini also generates Whatsup? no/creative_work, a span that does not appear in

the input text and matches no gold entity. In a generative setting, a semantically-aware metric should be able to tell apart correct extractions (i.e., aliases, paraphrases and omissions in the gold) from true hallucinations.

3 Related Work

Traditionally, NER has been evaluated with precision, recall, and F1 under exact span and label match, where a prediction is correct only if both its boundary and type are identical to the gold annotation (Chinchor, 1995; Tjong Kim Sang and De Meulder, 2003). SemEval-2013 Task 9.1 softened this setting by proposing multiple views of correctness, including strict, exact, partial, and type matching (Segura-Bedmar et al., 2013).

Noisy user-generated text. User-generated text has made these limitations especially visible (Baldwin et al., 2015; Strauss et al., 2016). Derczynski et al. (2017) shifted attention to novel and emerging entities, noting the recall difficulties caused by rare and unstable surface forms, while recent work continues this concern through noise analyses for NER (Bhadauria et al., 2024) and Large Language Model (LLM)-based extraction from informal text (Popovic et al., 2024). Our work follows this line, but treats noisy text as one instance and a symptom of a broader evaluation problem.

Relaxing exact scoring. Several lines of work relax exact scoring using external structures. Hierarchical precision/recall/F1 gives partial credit when predicted and gold labels are close in a taxonomy (Kiritchenko et al., 2006), while canonicalization approaches resolve entities to a common name through span-overlap scoring, type weights, entity linking, or coreference (Hachey et al., 2014; Agarwal et al., 2019; Bast et al., 2023). Yamada et al. (2015) and Limsopatham and Collier (2016) explored entity linking and neural sequence models. Generative linking systems such as mGENRE predict canonical entity names autoregressively, but require a Knowledge Base (KB) and a linkable target (De Cao et al., 2022). These approaches show that exact-match F1 is often too strict, but they typically require ontology-aware resources, alias tables, or a fixed KB, mitigating closed-world variation rather than addressing open-ended generative outputs.

LLM-based entity matching and linking. Recent work has applied generative LLMs to entity

matching and entity linking. Peeters and Bizer (2023); Peeters et al. (2025) show that GPT can perform competitive in zero-shot and few-shot entity matching with far fewer labeled examples than a fine-tuned encoder. Wang et al. (2025b) move beyond binary decisions by prompting LLMs to compare or select among multiple candidate records. For entity linking, Ding et al. (2024) constrain an LLM to choose among retrieved candidates, achieving comparable performance to supervised model. However, these methods are susceptible to LLM selection and prompt variation, leading to more instability during matching/linking.

Generative-based NER. Recent work in generative relation extraction argues that reference-based precision and recall are not sufficient when LLMs can produce valid outputs that differ from the annotation surface form. Jiang et al. (2024) make this point explicitly in GenRES, which augments standard scores with dimensions such as factualness and completeness. From a more general structured-prediction perspective, Chen et al. (2023) argue that evaluation design strongly shapes what systems are rewarded for producing. In generative NER specifically, systems such as GPT-NER reformulate extraction as text generation and highlight hallucination control as a central problem (Wang et al., 2025a). Recent surveys document the rapid adoption of generative Information Extraction (IE) pipelines across NER, relation extraction, and event extraction (Xu et al., 2024). More broadly, semantic generation metrics such as BERTScore (Zhang et al., 2020) show the value of replacing string match overlap with contextual similarity while keeping the precision/recall intuition intact. Our setting differs from pure free-form generation (e.g., Machine Translation), but the underlying evaluation problems are shared.

Annotation errors. A growing body of work shows that gold NER and IE annotations contain omissions and outright mistakes, which become more apparent as the model quality increases. For instance, Reiss et al. (2020) identify over 1.3k incorrect labels in CoNLL-2003 and note that the number of test-set errors is comparable to the remaining errors of strong models; Rueda et al. (2024) subsequently introduce CoNLL# as a corrected test set for lower-noise evaluation. For this reason, researchers treat partially annotated or unlabeled entities as a central learning problem rather than as negligible noise (Mayhew et al., 2019; Peng

et al., 2019; Li et al., 2021). Similar observations hold beyond NER: Alt et al. (2020) show that relation extraction performance on TACRED is substantially distorted by annotation error, and Nejadgholi et al. (2020) report that for overlapping same-label span mismatches in medical NER, expert users accept or partially accept the large majority of cases. This literature supports a key premise of our work: in tasks like generative NER, a false positive may be an annotation error or a model error, such as a hallucination, but it may also be an alternative mention or a contextually relevant prediction.

Our work is close in spirit to the semantic and canonicalization approaches, but differs in two ways. First, we target generative NER directly, where the output may be a plausible mention or type that is not ontology-normalized and may not overlap with the gold span. Second, instead of collapsing everything onto a single metric, we provide a separate metric that shows how many over-generated predictions are unsupported by the gold annotation. Like ontology-aware and linking-based approaches, **GeNERscore** credits semantic equivalence, but unlike them, it does not require a reference glossary or knowledge base.

4 GeNERscore

We describe **GeNERscore**, a semantically-aware evaluation framework for generative and open-vocabulary NER that operates on the embedding space rather than relying on string matching.

4.1 Motivation

Let $\bar{E} = \{\bar{e}_1, \dots, \bar{e}_N\}$ denote the gold typed entities and $\hat{E} = \{\hat{e}_1, \dots, \hat{e}_M\}$ the predicted typed entities. Under strict evaluation, a prediction is correct only if it exactly matches a gold entity in both mention and type. Writing $\mathbf{1}[\bar{e}_i = \hat{e}_j]$ for this exact-match indicator, the standard entity-level scores are as follows.

$$\text{Exact Precision} = \frac{1}{M} \sum_{j=1}^M \max_{i=1}^N \mathbf{1}[\bar{e}_i = \hat{e}_j], \quad (1)$$

$$\text{Exact Recall} = \frac{1}{N} \sum_{i=1}^N \max_{j=1}^M \mathbf{1}[\bar{e}_i = \hat{e}_j] \quad (2)$$

These metrics are easy to interpret and are applied when span-faithful extraction and type identity are required. However, they also come with an important consideration: a prediction either gets full credit or none.

One relaxation is partial matching, which could be implemented in multiple ways based on substring or span overlap. It gives credit to near misses due to boundary drift, while still limited to surface-form similarity. It helps with cases like *Apple* versus *Apple Inc.*, but not with aliases, descriptive mentions, or semantically equivalent references. In summary, partial matching softens exact scoring without distinguishing semantic recovery from partial overlap over-credit. Such considerations might not be ideal for generative and open-vocabulary NER, where the evaluation focuses on whether predicted typed entities belong to the same semantic frame defined by the domain of interest rather than on strict string matching (§2), including cases such as paraphrases, hypernyms, acronyms, and, more in general, semantic relatedness.

We design **GeNERscore** to account for such cases by satisfying the following desiderata via the replacement of string matching with embedding-based semantic similarity. First, it rewards semantic similarity beyond lexical identity, to accommodate aliases, abbreviations, and spelling variants. Second, it handles type mismatch beyond exact match, to also reward types’ semantic similarity. Third, it still penalizes incompatible types and wrong boundaries, such that mention or type overlap alone are not sufficient for correctness. Fourth, it exposes over-generated entities via a separate score.

4.2 Method

GeNERscore operates by computing the similarity matrix $S \in \mathbb{R}^{N \times M}$, where each entry S_{ij} quantifies how semantically close gold entity $\bar{e}_i$ is to predicted entity $\hat{e}_j$. Formally, we assume a pre-trained text embedding model $\phi : \text{String} \rightarrow \mathbb{R}^D$ that maps arbitrary strings to D -dimensional embeddings. We denote a typed entity e as a pair $e = (e^{\text{txt}}, e^{\text{typ}})$, where e^{txt} is the mention string and e^{typ} is the type label. **GeNERscore** is therefore entity-level: it compares entity strings, rather than considering per-token scoring. We do so to account for artifacts a generative model might produce, such as normalized, paraphrased, or descriptive mentions that are not a token-by-token copy of the input. Given embedding model ϕ and typed entity e , we define its embedding as follows:

$$\begin{aligned} \mathbf{e} &= (e^{\text{txt}}, e^{\text{typ}}), \\ e^{\text{txt}} &= \phi(e^{\text{txt}}) \in \mathbb{R}^D, \\ e^{\text{typ}} &= \phi(e^{\text{typ}}) \in \mathbb{R}^D \end{aligned} \quad (3)$$

We represent each typed entity e with separate mention and type embeddings. The mention embedding captures referential similarity between surface forms, while the type embedding captures compatibility between labels. For both embeddings, we embed the given raw string (e.g., PER for types). We apply no canonicalization or mapping to keep the metric ontology-agnostic: it requires no per-dataset label dictionary and allows for free-form or previously unseen labels.

Conceptually, S is an entity-level analogue of semantic generation metrics such as BERTScore (Zhang et al., 2020). The main difference is that the pairwise comparison is between typed spans rather than tokens.

Type-aware Cosine Similarity. Given gold and predicted entity embeddings, we define S by computing a similarity function $s(\cdot, \cdot)$ of each $(\bar{e}, \hat{e})$ pair as follows.

$$S_{ij} = s(\bar{e}_i^{\text{txt}}, \hat{e}_j^{\text{txt}}) \cdot s(\bar{e}_i^{\text{typ}}, \hat{e}_j^{\text{typ}}) \quad (4)$$

where

$$s(\bar{e}_i^{\text{txt}}, \hat{e}_j^{\text{txt}}) = \max(0, \cos_sim(\bar{e}_i^{\text{txt}}, \hat{e}_j^{\text{txt}})), \quad (5)$$

$$s(\bar{e}_i^{\text{typ}}, \hat{e}_j^{\text{typ}}) = \max(0, \cos_sim(\bar{e}_i^{\text{typ}}, \hat{e}_j^{\text{typ}})) \quad (6)$$

We keep the two comparisons separate and combine them multiplicatively so that credit requires *joint* compatibility: a low score in either embedding suppresses the product, whereas a sum or average would let a strong mention match mask an incompatible type. Each similarity is clipped at zero before being multiplied, so a negative (incompatible) mention or type similarity contributes no credit, rather than flipping the sign of the product. This bounds every entry $S_{ij} \in [0, 1]$ and keeps the score monotone in both embeddings.

Interpretation. Each row i of S shows how well gold entity $\bar{e}_i$ is matched by every prediction; the row maximum identifies its best candidate recovery. Each column j shows how well the predicted entity $\hat{e}_j$ aligns with gold annotations; the column maximum indicates its strongest semantic support. High values suggest referential and type compatibility, whereas uniformly low columns indicate likely unsupported predictions. This row/column duality allows S to expose entity-level alignment and to support both recall-like and precision-like views.

4.3 Assignment

The similarity matrix alone does not specify which predictions should count as matches for gold entities. We therefore define an assignment set $\mathcal{A}$ over pairs (i, j) and discuss three algorithms to define $\mathcal{A}$.

4.3.1 Hungarian Assignment

The Hungarian algorithm produces a globally optimal one-to-one alignment in polynomial time. (Kuhn, 1955) Polynomial complexity is important as each matrix S instance is composed of N rows and M columns, thus potentially quadratic. Finding the optimal alignment in time may be difficult for large documents. The Hungarian algorithm solves this issue.

$$\mathcal{A}_H = \arg \max_{\mathcal{A}} \sum_{(i,j) \in \mathcal{A}} S_{ij} \quad (7)$$

subject to the constraint that each gold entity and each prediction participates in at most one pair. This provides a maximum-weight matching, but it cannot express cases where one entity is split across several predicted mentions. It is therefore too restrictive as a standardized assignment rule for generative NER: aliases, expanded descriptions, split and merged entities may produce many-to-many alignments.

4.3.2 Threshold Assignment

Threshold assignment relaxes one-to-one matching by selecting every pair above a user-specified threshold $\tau \in [0, 1]$.

$$\mathcal{A}_\tau = \{(i, j) : S_{ij} \geq \tau\} \quad (8)$$

This many-to-many strategy is better suited to boundary ambiguity and split or merged predictions, but it depends directly on threshold calibration. As τ rises, low-confidence but still best-available correspondences are removed entirely. This can lower the recall-oriented score even when every gold entity has a plausible best prediction, because threshold assignment does not guarantee that each gold item receives any partner.

4.3.3 Hybrid Assignment

A hybrid strategy that combines the Hungarian assignment with threshold-based expansion as follows.

$$\mathcal{A}_{\text{hybrid}} = \mathcal{A}_H \cup \mathcal{A}_\tau \quad (9)$$

The result preserves the best one-to-one backbone while still admitting additional semantically plausible matches. The Hungarian component guarantees entity coverage up to the available number of predictions, while the threshold component maintains the many-to-many assignment needed for the reasons discussed above.

4.4 Evaluation Metrics

Given S and an assignment $\mathcal{A}$, **GeNERscore** comprises three complementary metrics. The first two are soft analogues of recall and precision. The third measures how much of a model’s prediction is not supported by gold entities.

4.4.1 Soft Recall

For each gold entity $\bar{e}_i$, we take the best similarity among all predictions assigned to it, or 0 if it is unmatched.

$$r_i = \begin{cases} \max_{j:(i,j) \in \mathcal{A}} S_{ij} & \text{if } \exists j \text{ s.t. } (i, j) \in \mathcal{A}, \\ 0 & \text{otherwise} \end{cases} \quad (10)$$

We then define soft recall as follows.

$$\text{Soft Recall} = \frac{1}{N} \sum_{i=1}^N r_i \quad (11)$$

While there may be multiple assigned predictions to a gold entity, we take the maximum similarity over assigned predictions rather than their mean. A model that generates several plausible aliases of one gold entity (e.g., *Apple*, *Apple Inc.*) is therefore not heavily penalized for possible redundancies in generated entities.

4.4.2 Soft Precision

For each prediction $\hat{e}_j$, we take the mean similarity across its assigned gold assignments when it is matched.¹ Otherwise, in the case $\hat{e}_j$ is unmatched, we consider the best similarity score across gold annotations.

$$c_j = \begin{cases} \frac{1}{|C_j|} \sum_{i:(i,j) \in \mathcal{A}} S_{ij} & \text{if } C_j \neq \emptyset, \\ \max_{i=1}^N S_{ij} & \text{otherwise} \end{cases} \quad (12)$$

where $C_j = \{i : (i, j) \in \mathcal{A}\}$ is the set of gold entities indexes assigned to prediction j . We define

¹For one-to-one assignment, this reduces to the single assignment score.

soft precision as follows.

$$\text{Soft Precision} = \frac{1}{M} \sum_{j=1}^M c_j \quad (13)$$

We average over assigned predictions to account for the case where a prediction might be matched to more than one gold entity. By doing so, we avoid crediting a prediction solely for its best match when considering a many-to-many assignment. This design finds its motivation when considering the case of over-generation, which occurs when a model generates more entities $\hat{e}$ than the available gold annotations $\bar{e}$ (i.e., $M > N$). In such a case, soft precision weights surplus predictions proportionally to their similarity scores rather than attributing a binary penalization (i.e., either correct or incorrect). For instance, a surplus mention that is a plausible alias of a gold entity is penalized only lightly. Instead, an unmatched mention with low similarity score lowers the overall precision score.

4.4.3 Unsupported Rate

The unsupported rate quantifies the proportion of predictions that are distant from the gold entities, according to some similarity threshold. It is assignment-independent. For each prediction, we compute its best similarity to any gold entity (i.e., its *support*).

$$h_j = \max_{i=1}^N S_{ij} \quad (14)$$

Given a support threshold τ_u , we flag predictions with insufficient support as unsupported.

$$\text{Unsupported Rate} = \frac{1}{M} \sum_{j=1}^M \mathbf{1}[h_j < \tau_u] \quad (15)$$

It is worthwhile noticing that an unmatched prediction may be the product of a genuine hallucination or of a mention that the model correctly emits but the gold annotation fails to specify (§2). This metric should therefore be used as a way to estimate the ratio of the prediction that is over-generated or not correspondent to any gold, rather than a hallucination gauge.

5 Experimental Setup

To validate our evaluation framework, we consider two settings. In the first, we assess whether semantic scoring agrees with existing scoring methods on standard NER benchmarks, where supervised

discriminative models achieve near-optimal performance according to exact-match scoring. As benchmarks, we consider CoNLL-2003 (Tjong Kim Sang and De Meulder, 2003) and OntoNotes 5 (Weischedel et al., 2013). In the second setting, we stress-test the metrics to expose different kinds of divergence, seen in §2, between strict string matching and semantically-aware scoring. As benchmarks, we consider TweetNER7 (Ushio et al., 2022), a noisy dataset we deem valuable for assessing aliasing, abbreviation, annotation omission, boundary variation, and emerging surface forms. Additionally, we consider SciERC (Luan et al., 2018), a scientific-domain dataset where annotations contain specialized terminology and annotation disagreement often concerns type mismatch and technical paraphrasing rather than informal spelling or event-driven naming.

Models. We consider three model architectures: T-NER (Ushio and Camacho-Collados, 2021), GPT 5.4-mini (Singh et al., 2025)² and GLiNER2 (Zaratiana et al., 2025)³. T-NER represents task-specific token-classification systems comprising a transformer encoder and Conditional Random Field (CRF)-style decoding. We use the existing T-NER model checkpoints for CoNLL-2003⁴, OntoNotes 5⁵ and TweetNER7⁶. For SciERC, we train a ModernBERT (Warner et al., 2025) encoder using T-NER implementation with default hyperparameters on the training set. T-NER represents a strong baseline when trained for a target dataset. Nonetheless, the model requires re-training when the domain or the set of entity types changes. GPT 5.4-mini represents instruction-following generative extraction. It can be prompted with a new set of entity types without task-specific fine-tuning. For each dataset, we zero-shot prompt GPT 5.4-mini with the dataset’s allowed entity types to return a JSON array of {text, type} objects, which we parse into typed entity spans (see Appendix A). GLiNER2 represents a middle ground between fixed token classifiers and fully generative LLMs: it supports open-vocabulary entity types through natural-language labels with an encoder model backbone.

²Accessed via the OpenAI API (GPT-5.4-mini); see <https://platform.openai.com/docs/models>, June 2026.

³gliner2-large-v1.

⁴deberta-v3-large-conll2003.

⁵roberta-large-ontonotes5.

⁶roberta-large-tweetner7-all.

Model	Strict F1 ↑	Partial F1 ↑	Soft F1 ↑	U. Rate ↓
<i>CoNLL-2003</i> ($n = 3453$)				
T-NER	0.922	0.933	0.945	0.030
GLiNER2	0.355	0.380	0.592	0.470
GPT-5.4-mini	0.750	0.768	0.833	0.045
<i>OntoNotes 5</i> ($n = 8262$)				
T-NER	0.931	0.952	0.955	0.028
GLiNER2	0.348	0.418	0.533	0.577
GPT-5.4-mini	0.775	0.852	0.864	0.052
<i>TweetNER7</i> ($n = 2807$)				
T-NER	0.637	0.693	0.818	0.151
GLiNER2	0.263	0.324	0.581	0.516
GPT-5.4-mini	0.320	0.586	0.708	0.285
<i>SciERC</i> ($n = 100$)				
T-NER	0.505	0.650	0.744	0.117
GLiNER2	0.326	0.449	0.669	0.302
GPT-5.4-mini	0.324	0.422	0.640	0.283

Table 1: Results across all four datasets, macro-averaged over instances.

Evaluation. We compute **GeNERscore** metrics after parsing a model’s predictions into typed entity spans. For each dataset instance with gold entity types $\bar{E}$ and predicted entity types $\hat{E}$, we perform experiments with each assignment method and different thresholds. For each experiment, we compute the similarity matrix S , the assignment $\mathcal{A}$, and the metrics. We macro-average metrics over dataset instances in a given split. We supply GPT-5.4-mini and GLiNER2 with a dataset’s entity type ontology, such that gold and predicted entity types are drawn from the same vocabulary. We apply no further canonicalization to either type or mention strings before embedding them. We compute **GeNERscore** metrics using the embedding models and thresholds reported in Appendix B. We define partial matching in the following way: $\bar{e}^{txt}$ is a substring of $\hat{e}^{txt}$ or $\hat{e}^{txt}$ is a substring of $\bar{e}^{txt}$ and $\bar{e}^{typ} = \hat{e}^{typ}$. This corresponds to partial+type of Segura-Bedmar et al. (2013). Due to space limitations, we do not report all the data and figures related to all alignment algorithms, models and thresholds, but focus only on the most significant configurations.

6 Results

We perform an initial analysis on all configurations (Appendix C). We observe that the Hungarian assignment is too limiting, and threshold assignment can be too sensitive to the calibration of τ . Therefore, we focus on hybrid assignment. We also observe both in $\mathcal{A}_\tau$ and $\mathcal{A}_{hybrid}$ (§4), that *soft precision* and *soft recall* tend to meet around $\tau = 0.6$. Therefore, in Table 1, we report results using $\mathcal{A}_{hybrid}$, $\tau = \tau_u = 0.6$ and $\phi = \text{Qwen3-}$

embedding-4B as the embedding model, since it is the largest one in our experiments. For brevity, we summarize soft metrics with their harmonic mean, denoted as *Soft F1*. We observe two patterns. First, the discriminative T-NER models are the best-performing ones on all four datasets under both strict and **GeNERscore** evaluation. This demonstrates that our soft variants do not disagree with traditional metrics on discriminative models. Second, the gap between Strict F1 and Soft F1 is especially large for GPT 5.4-mini on a noisy benchmark such as TweetNER7. In this setting, the difference between GPT 5.4-mini’s and T-NER’s Soft F1 is much less pronounced than the difference between their Strict F1 scores. Both GPT 5.4-mini and GLiNER2 over-generate entities, yet both gain from semantic scoring: their Soft F1 exceeds their Partial F1 on every dataset. What separates them is the Unsupported Rate. GLiNER2 leaves roughly half of its predictions unsupported on CoNLL-2003, OntoNotes 5, and TweetNER7, which hampers its Soft F1. The two models converge on SciERC, where GLiNER2’s unsupported rate drops to 0.302 and its Soft F1 (0.669) slightly exceeds that of GPT 5.4-mini (0.640).

SciERC is the most challenging dataset in our experiments. In Figure 1, we report the different behaviors of our metrics when varying the assignment algorithm and thresholds. Hungarian assignment yields fixed soft precision and soft recall regardless of τ , providing the best one-to-one assignment. Threshold assignment allows calibration, but soft precision and soft recall diverge at the extremes of τ range. Hybrid assignment combines the stability of optimal matching with threshold-based filtering. We observe precision and recall curves cross near $\tau = 0.6$. The same pattern holds on the remaining datasets (Appendix C). We also verify that the choice of embedding model has little effect on the metrics (Appendix B).

7 Discussion

On CoNLL-2003 and OntoNotes 5, the supervised T-NER models have near-ceiling performance and are the best-performing models. **GeNERscore** does not over-allocate credit where exact match already works. Regarding TweetNER7 and SciERC, model performance drops compared to standard benchmarks and the strict-to-soft gap widens, most prominently for the generative model. Performance drop could be due to the fact that standard bench-

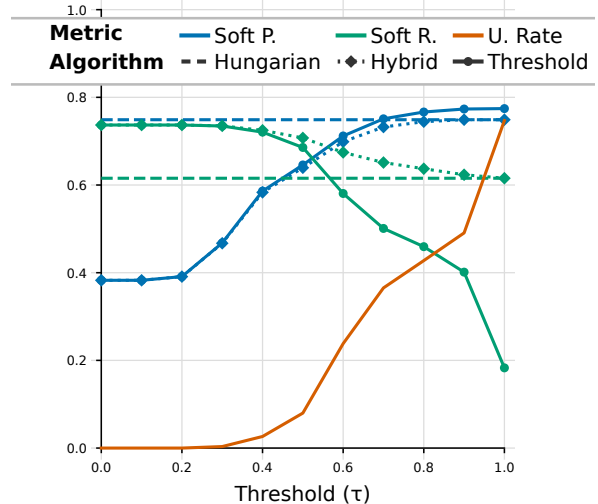


Figure 1: **GeNERscore** metrics for SciERC. Results averaged across models (T-NER, GLiNER2, GPT 5.4-mini). Thresholds $\tau = \tau_u$ for the three assignment algorithms. Hungarian assignment (dashed) is τ -independent. The unsupported rate is assignment-independent and appears as a single orange curve.

marks are simple and well-curated, while TweetNER7 adds noise and SciERC adds document context and specialized terms. Another hypothesis is test-set leakage into the backbones’ pretraining data.

The gap in the noisier data is what the metric is meant to catch. GPT 5.4-mini produces aliases, paraphrases, and near-synonymous type labels that strict matching scores as wrong. Soft scoring recovers this credit, while the unsupported rate stays above the supervised model’s, so **GeNERscore** does not reward all predictions. Soft scoring does not replace exact and partial matching, which remain the right choice when exact mention and type identity are important. **GeNERscore** is complementary and most useful for generative and open-vocabulary systems where string matching is less reliable.

8 Conclusion

Exact-match NER evaluation is well suited to closed, span-faithful extraction, but it becomes increasingly brittle when model outputs and annotations diverge syntactically. We have shown six cases in which this evaluation falls short: alias and surface-form variation, type disagreement, semantic paraphrase, partial-overlap over-credit, annotation incompleteness and hallucination. **GeNERscore** is a modular framework that evaluates NER based on the semantic alignment between predicted

and gold typed entities.

We validate the framework against four datasets and three model families. We show that it does not disagree with traditional metrics on discriminative models. On generative models, it recovers credit that existing string-matching metrics withhold. Importantly, **GeNERscore** flags over-generation through a new metric called *unsupervised rate*. We show that our metrics are largely insensitive to the embedding model used to compute them, which makes them computationally viable. By agreeing with traditional metrics in cases where they are reliable and capturing edge cases where they struggle, **GeNERscore** enables a more semantically sensitive evaluation of generative NER systems.

Limitations

Soft precision credits a surplus prediction by its best available support, so it is lenient toward over-generation that stays close to the gold. That is what we want for generative output. When a stricter evaluation is needed, employing traditional precision is still advisable.

While we show that our metrics are robust with respect to the choice of embedding models (Appendix B), they inherit the limitations and biases of the underlying embedding model.

All our metrics rest on the assumption that a high alignment score reflects referential and type similarity. A human alignment study would test this directly. As this would entail a significant amount of work, we leave it to future works, alongside testing datasets in other languages and domains.

While LLM-based methods come with different biases and suffer from prompt sensitivity, they can provide reasoning behind a given match. A comparison with them on entity matching and entity linking could help in untangling the factors described above.

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⁷<https://decisionbrain.com>

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A Generation and Parsing

We aggregate T-NER output to transform sequences of tokens in spans, GLiNER2 uses its native span output; only GPT 5.4-mini needs a prompt and parser.

Prompt. We query GPT 5.4-mini zero-shot with the template below. `{entity_types}` is the dataset’s allowed types as a comma-separated list, and `{text}` the input sentence. We ask the model to produce exact mentions and use the provided types to put it in the best conditions for this task.

```
You are a Named Entity Recognition (NER)
system. Your task is to extract all named
entities from the given text.
Allowed entity types: {entity_types}
Text: "{text}"

Return ONLY a JSON array of objects
matching this schema:
[{"text": "exact entity mention from
the text", "type": "one of the allowed
entity types"}]

Rules:
- If no entities are found, return an
empty array: []
- Do not include explanations, only
output the JSON array.
```

Hyperparameters. The following hyperparameters were used: `top_p = 1.0`, `max_tokens = 512`, `seed = 42`. For GLiNER2, we used `gliner_threshold = 0.5`.

Parsing. We parse the response as a JSON array of `{text, type}` objects. Strings are kept verbatim; unparseable responses give an empty prediction set.

B Robustness to Embedding Model

We study how **GeNERscore** and the metrics vary as the embedding model varies. We consider the following models selected from the MTEB/MMTEB leaderboards (Muennighoff et al., 2023; Enevoldsen et al., 2025): Qwen3-Embedding-4B (Zhang et al., 2025) as a representative of big models, Qwen3-Embedding-0.6B (Zhang et al., 2025) as a smaller version of the previous one; EmbeddingGemma-300M (Vera et al., 2025), jina-embeddings-v5-text-small (677M) and jina-embeddings-v5-text-nano (239M) (Akram et al., 2026) as smaller models.

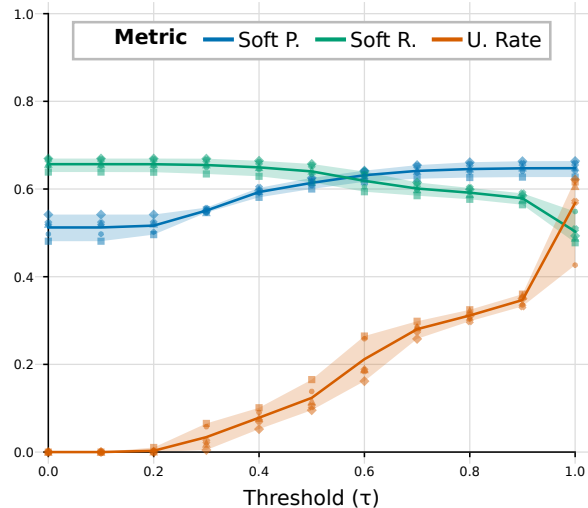


Figure 2: Robustness of the semantic metrics to the embedding model and change of τ/τ_u . Each line is the mean across five encoders and the band their min-max across all datasets and algorithms. For each fixed threshold, the metrics vary little across encoders over the whole τ range. Markers: ■ EmbeddingGemma 300M, ◆ Qwen3-Embedding 0.6B, ▲ Qwen3-Embedding 4B, ✖ Jina v5 Nano, ★ Jina v5 Small.

The choice of encoder has little effect on the metrics. Across five embedding models, *soft precision*, *soft recall*, and the *unsupported rate* stay within a narrow band over the entire τ sweep (Figure 2).

C Assignment Algorithms results

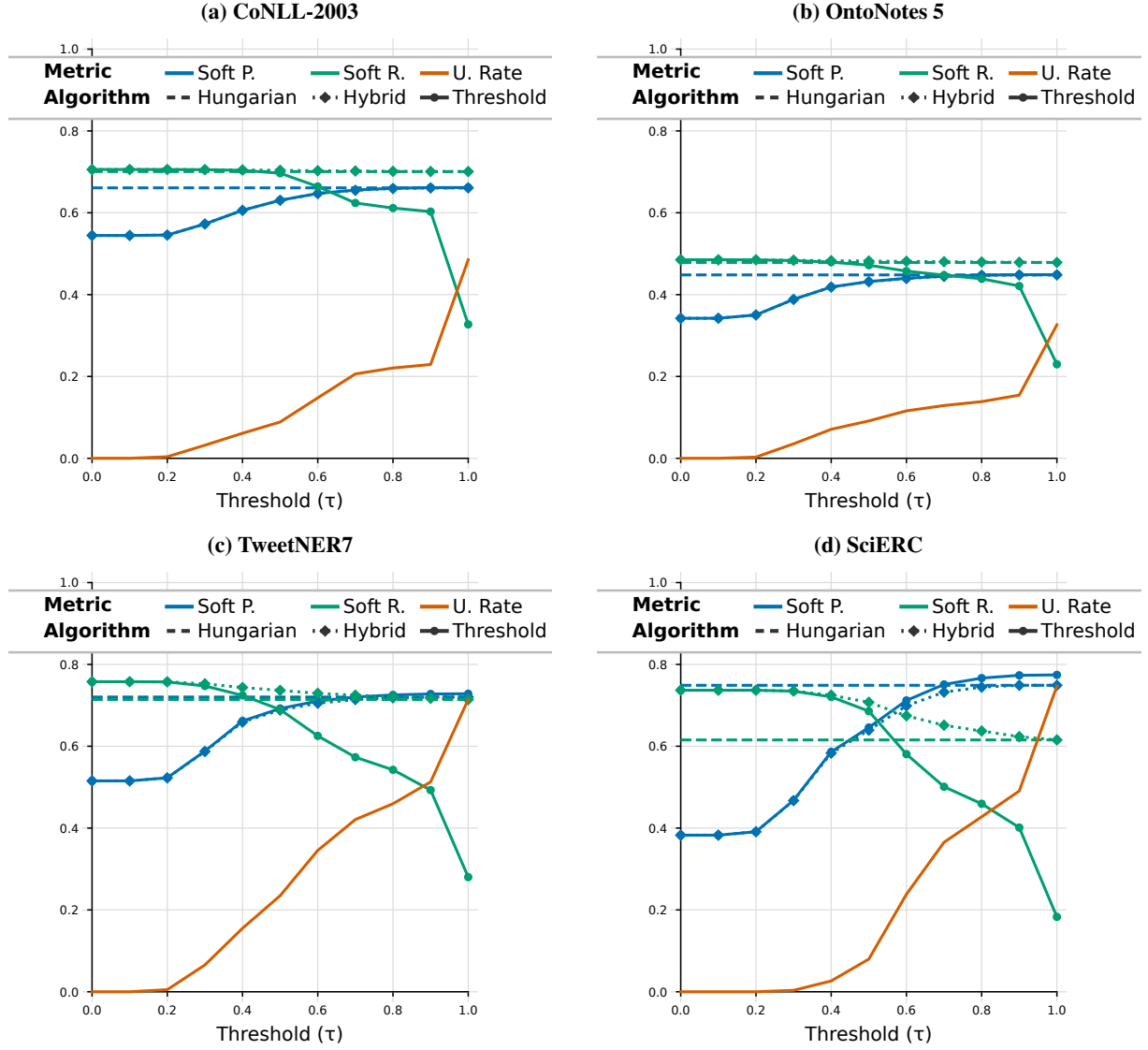


Figure 3: Soft precision (Soft P.), soft recall (Soft R.), and unsupported rate (U. Rate) as a function of $\tau = \tau_u$ for the three assignment algorithms, macro-averaged over T-NER, GLiNER2, and GPT 5.4-mini. Averaged across all embedding models (Appendix B). Hungarian assignment (dashed) is τ -independent. The unsupported rate is assignment-independent and appears as a single orange curve per panel.

D Further Examples

We report 30 further examples from TweetNER7 and SciERC where strict evaluation fails to account for meaning equivalence. For each sample we show gold entities and per-model predictions side by side, with strict F_1 , Soft F_1 , and unsupported rate U ($\tau=\tau_u=0.6$, hybrid assignment, Qwen3-Embedding-0.6B). Each example is tagged with the phenomena it exhibits (§2). Type abbreviations: per=person, loc=location, grp=group, corp=corporation, evt=event, cw=creative_work, prod=product; for SciERC: Meth=Method, Mat=Material, Gen=Generic, Met=Metric, OST=OtherScientificTerm. Characters that cannot be rendered in this document are replaced with descriptive placeholders: [emoji] for emoji, [Thai] for Thai script, and [Cyrillic] for Cyrillic script.

TweetNER7

TweetNER7 #2551

Alias · Partial · Omiss.

“snl had the PERFECT opportunity to do a “ target lady during covid ” sketch with kristen wiig and chose not too ??? yeah, I need answers { @Saturday Night Live - SNL@ }”

Gold	T-NER	GLiNER2	GPT-5.4-mini
sketch/prod target/corp { @Saturday Night Live - SNL@ }/prod	kristen wiig/per snl/grp { @Saturday Night Live - SNL@ }/prod	SNL/corp kristen wiig/per sketch/cw	Kristen Wiig/per SNL/corp Saturday Night Live/cw target lady during covid/cw
	F_1 .333 $\hat{F}_1$.434 U .667	F_1 .000 $\hat{F}_1$.332 U .667	F_1 .000 $\hat{F}_1$.293 U .750

TweetNER7 #1684

Alias · Paraph. · Partial · Omiss.

“Severe flooding due to Tropical Storm Ulysses Typhoon Vamco Philippines ... { {URL} } via { @YouTube@ } # UlyssesPH # PhilippinesNeedsHelp # Philippines # GlobalWarming # Vamco # storm # TropicalCyclone”

Gold	T-NER	GLiNER2	GPT-5.4-mini
Philippines/loc Tropical Storm Ulysses/evt { @YouTube@ }/prod	Philippines/loc Tropical Storm Ulysses Typhoon Vamco/evt { @YouTube@ }/prod	GlobalWarming/evt Philippines/loc TropicalCyclone/evt Typhoon Vamco/evt Vamco/evt storm/evt	Philippines/loc PhilippinesNeedsHelp/evt Tropical Storm Ulysses/evt TropicalCyclone/evt Typhoon Vamco/evt UlyssesPH/evt Vamco/evt YouTube/corp
	F_1 .667 $\hat{F}_1$.972 U .000	F_1 .222 $\hat{F}_1$.692 U .167	F_1 .364 $\hat{F}_1$.719 U .250

GPT predicts “Typhoon Vamco” for gold “Tropical Storm Ulysses: the same storm under its international versus Philippine-specific name. A cross-lingual alias that only embedding-based scoring can capture.

TweetNER7 #3

Alias · Type · Omiss.

“- Albin Ekdal is the 1st player from Sweden to score a Serie A goal for { @U.C. Sampdoria@ } since Lennart Skoglund on 24 September 1961, also against AC Milan (3-2 away win). # SerieATIM # SampdoriaMilan”

Gold	T-NER	GLiNER2	GPT-5.4-mini
AC Milan/grp Albin Ekdal/per Lennart Skoglund/per Sweden/loc { @U.C. Sampdoria@ }/grp	AC Milan/grp Albin Ekdal/per Lennart Skoglund/per SampdoriaMilan/grp Serie A/evt Sweden/loc { @U.C. Sampdoria@ }/grp	AC Milan/corp Albin Ekdal/per Lennart Skoglund/per Serie A/evt Sweden/loc	24 September 1961/evt AC Milan/corp Albin Ekdal/per Lennart Skoglund/per SampdoriaMilan/evt Serie A/cw SerieATIM/evt Sweden/loc U.C. Sampdoria/corp
	F_1 .833 $\hat{F}_1$.936 U .143	F_1 .600 $\hat{F}_1$.782 U .200	F_1 .429 $\hat{F}_1$.768 U .333

All models identify the core entities but disagree on whether sports teams are groups or corporations; the gold also omits in-text mentions such as “Serie A,” penalizing models that extract them.

TweetNER7 #43

Type · Omiss.

“I know everyone loves Fauci and govt agencies make mistakes but I really think the CDC needs an overhaul. Needs better leadership,[@Joe Biden@]”

Gold	T-NER	GLiNER2	GPT-5.4-mini
CDC/corp {@Joe Biden@}/per	CDC/corp Fauci/per {@Joe Biden@}/per	CDC/corp Fauci/corp govt agencies/corp	CDC/grp Fauci/per Joe Biden/per
	F_1 .800 $\hat{F}_1$.921 U .333	F_1 .400 $\hat{F}_1$.600 U .000	F_1 .000 $\hat{F}_1$.752 U .333

TweetNER7 #2364

Alias · Partial · Omiss.

“Everybody is picking the Cowboys,on Gameday shows. Lets prove them wrong. LETS GO BIG BLUE. {@LPG - NYG@} # NYG # NYGvsDAL Andy Dalton talking about watching the Washington game after this game. He thinks they have already won # GiantsPride # Giants”

Gold	T-NER	GLiNER2	GPT-5.4-mini
Cowboys/grp Giants/grp GiantsPride/grp NYG/grp NYGvsDAL Andy Dalton/evt Washington game/evt {@LPG - NYG@}/per	Andy Dalton/per Cowboys/grp Giants/grp NYG/grp NYGvsDAL/evt Washington/grp {@LPG - NYG@}/grp	Andy Dalton/per Cowboys/grp Washington/loc Washington game/evt	Andy Dalton/per BIG BLUE/grp Cowboys/grp Gameday/cw Giants/grp GiantsPride/grp NYG/grp NYGvsDAL/evt Washington/grp
	F_1 .429 $\hat{F}_1$.805 U .143	F_1 .364 $\hat{F}_1$.703 U .500	F_1 .500 $\hat{F}_1$.761 U .222

“BIG BLUE” is a well-known nickname for the New York Giants, an alias that would be penalized even if it was annotated in the gold.

TweetNER7 #2414

Alias · Type · Halluc.

“Jill Biden Leaves the White House in Green Jewel-Toned Jacket & Her Favorite Boot Trend {{URL}} via {@Yahoo@} Hey {@MELANIA TRUMP@},did you see what Dr. Biden was wearing the other day ?”

Gold	T-NER	GLiNER2	GPT-5.4-mini
Boot Trend/prod Dr. Biden/per Green Jewel-Toned Jacket/prod Jill Biden/per White House/loc {@MELANIA TRUMP@}/per {@Yahoo@}/prod	Dr. Biden/per Jill Biden/per White House/loc {@MELANIA TRUMP@}/per {@Yahoo@}/prod	Boot Trend/prod Dr. Biden/per Green Jewel-Toned Jacket/prod Jill Biden/per White House/loc	Dr. Biden/per Jill Biden/per MELANIA TRUMP/per Whatsup? no/cw White House/loc Yahoo/corp
	F_1 .833 $\hat{F}_1$.807 U .000	F_1 .833 $\hat{F}_1$.888 U .000	F_1 .462 $\hat{F}_1$.648 U .333

GPT hallucinates “Whatsup? no” as an entity while also producing the valid alias “MELANIA TRUMP.” “Yahoo” as corporation versus gold product shows type ambiguity. Valid and invalid predictions coexist.

TweetNER7 #307

Alias · Paraph. · Omiss.

“Okay. I’ve been a mess all over the weekend because of these award shows. Will watch the replay tom. Congrats,our love {@BTS_twt@} u deserve all of it. I love every single one of you. Yoongi baby,you can finally get some good rest. My 6 tannies,please be safe .”

Gold	T-NER	GLiNER2	GPT-5.4-mini
Yoongi/per {@BTS_twt@}/grp	Yoongi/per {@BTS_twt@}/grp	Yoongi baby/per award shows/evt tom/per	BTS_twt/grp Yoongi/per tannies/grp
	F_1 1.00 $\hat{F}_1$ 1.00 U .000	F_1 .000 $\hat{F}_1$.518 U .333	F_1 .400 $\hat{F}_1$.845 U .333

GPT predicts “tannies,” a fan nickname for BTS that is semantically plausible but absent from gold. Strict matching penalizes the model for capturing world knowledge the annotation misses.

TweetNER7 #782 Type
“Calpulli Mexican Dance Company presents “Dia de los Muertos,” a universal love story that captures the sentiment,color,and hope of day of the dead through music and dance. #DiaDeMuertos #DayoftheDead {{USERNAME}} {{URL}}”

Gold	T-NER	GLiNER2	GPT-5.4-mini
Calpulli Mexican Dance Company/grp DayoftheDead/evt Dia de los Muertos/cw DiaDeMuertos/evt	Calpulli Mexican Dance Company/grp DayoftheDead/evt Dia de los Muertos/evt DiaDeMuertos/evt	Calpulli Mexican Dance Company/corp Dia de los Muertos/evt dance/cw day of the dead/evt music/cw	Calpulli Mexican Dance Company/corp DayoftheDead/cw Dia de los Muertos/cw
F_1 .750 $\hat{F}_1$.910 U .000 F_1 .000 $\hat{F}_1$.732 U .400 F_1 .286 $\hat{F}_1$.675 U .000			

TweetNER7 #815 Type · Paraph. · Omiss.
“Watching Tokyo 7th Sister live with {{USERNAME}} and {{USERNAME}} and T7S is lit. Watching Chain The Blossom live,and I really to like to see them live in person soon when I can. I miss eventing in general,but T7S is on my list to see now with friends.”

Gold	T-NER	GLiNER2	GPT-5.4-mini
Chain The Blossom/evt Tokyo 7th Sister/evt	Chain The Blossom/grp T7S/grp Tokyo 7th Sister/grp	T7S/corp T7S/per Tokyo 7th Sister live/evt eventing/evt friends/grp {{USERNAME}}/per	Chain The Blossom/cw T7S/cw Tokyo 7th Sister/cw
F_1 .000 $\hat{F}_1$.648 U .333 F_1 .000 $\hat{F}_1$.491 U .667 F_1 .000 $\hat{F}_1$.582 U .333			

GPT predicts “T7S,” a standard abbreviation of the gold entity “Tokyo 7th Sister.” Strict matching rejects the alias entirely, soft scoring recovers it.

TweetNER7 #1002 Alias
““ Living is n’t easy but if you can manage to look past your own fears and everyone else ’s expectations,the reward is your life. “ Live in the moment. Love every moment. The message in {@Jaden@} Smith ’s latest movie that we could all benefit from. {{URL}}”

Gold	T-NER	GLiNER2	GPT-5.4-mini
{{@Jaden@}}/per	{{@Jaden@}}/per	movie/cw	Jaden Smith/per
F_1 1.00 $\hat{F}_1$ 1.00 U .000 F_1 .000 $\hat{F}_1$.000 U 1.00 F_1 .000 $\hat{F}_1$.601 U .000			

TweetNER7 #1230 Alias · Partial · Halluc.
“# MMATwitter poll tournament Finale {{USERNAME}} Vs {@C H[emoji]T O[emoji]V E R A@} Vs {@Chan Sung Jung@} Vs {@Kevin Holland@} Vote however you like it does n’t have to be who wins the fight it can be who you like more and so forth”

Gold	T-NER	GLiNER2	GPT-5.4-mini
MMATwitter poll tournament Finale/evt {@C H[emoji]T O[emoji]V E R A@}/per {@Chan Sung Jung@}/per {@Kevin Holland@}/per	MMATwitter/evt tournament Finale/evt {@C H[emoji]T O[emoji]V E R A@}/per {@Chan Sung Jung@}/per {@Kevin Holland@}/per	USERNAME/per fight/evt tournament Finale/evt	Chan Sung Jung/per Kevin Holland/per MMATwitter/grp Whatsup/per
F_1 .667 $\hat{F}_1$.852 U .000 F_1 .000 $\hat{F}_1$.249 U .667 F_1 .000 $\hat{F}_1$.622 U .250			

A mix of valid aliases (“Chan Sung Jung,” “Kevin Holland” stripped of markup) and a genuine hallucination (“Whatsup” as a person). Soft scoring rewards the former while the unsupported rate flags the latter.

TweetNER7 #1266 Alias
“i have a really good feeling the trailer is coming out in like 5 minutes since it ’s gonna be 12 please disney {@Disney TVA@}”

Gold	T-NER	GLiNER2	GPT-5.4-mini
{{@Disney TVA@}}/corp	{{@Disney TVA@}}/prod	disney/corp trailer/cw	Disney TVA/corp disney/corp
F_1 .000 $\hat{F}_1$.654 U .000 F_1 .000 $\hat{F}_1$.683 U .500 F_1 .000 $\hat{F}_1$.856 U .000			

TweetNER7 #1284

Alias

“Please please please,next week re shuffle the guys. . too boring #TV3GH #DateRush {@G i o v a n i CALEB #DATErush@}”

Gold	T-NER	GLiNER2	GPT-5.4-mini
{@G i o v a n i CALEB #DATErush@}/per	{@G i o v a n i CALEB #DATErush@}/per	CALEB/per DateRush/evt TV3GH/prod guys/grp	DATErush/cw DateRush/cw Giovani Caleb/per TV3GH/corp
	F_1 1.00 $\hat{F}_1$ 1.00 U .000	F_1 .000 $\hat{F}_1$.570 U .750	F_1 .000 $\hat{F}_1$.570 U .750

TweetNER7 #1350

Alias · Type · Partial · Omiss.

“Excited to be on the beat for tonight ’s Rangers-Devils tilt The first of four games between this team this week from the Rock begins at 7 Nico Hischier returns to the lineup for NJ {{USERNAME}} #NYR #NJDevils”

Gold	T-NER	GLiNER2	GPT-5.4-mini
NJDevils/cw Nico/loc Rangers-Devils/evt games/grp	NJ/grp NJDevils/grp NYR/grp Nico Hischier/per Rangers-Devils tilt/evt	NJ/corp Nico Hischier/per	Devils/grp NJ/loc NJDevils/grp NYR/grp Nico Hischier/per Rangers/grp the Rock/loc
	F_1 .000 $\hat{F}_1$.602 U .400	F_1 .000 $\hat{F}_1$.000 U 1.00	F_1 .000 $\hat{F}_1$.646 U .143

“Devils” is a valid short form of “NJDevils,” and “NJDevils” is typed as “group” versus gold “creative_work.”

TweetNER7 #1467

Paraph. · Partial

“< strong >26 Ways I Can Do More to Support the Black Lives Matter Movement </ strong >{{URL}} via {{USERNAME}}”

Gold	T-NER	GLiNER2	GPT-5.4-mini
Black Lives Matter/grp	Black Lives Matter Movement/grp	Black Lives Matter Movement/grp USERNAME/per	26 Ways I Can Do More to Support the Black Lives Matter Movement/cw Black Lives Matter Movement/grp
	F_1 .000 $\hat{F}_1$.967 U .000	F_1 .000 $\hat{F}_1$.775 U .500	F_1 .000 $\hat{F}_1$.840 U .500

“Black Lives Matter Movement” is an expansion of the gold entity “Black Lives Matter.” Soft scoring correctly rewards this near-match that strict matching rejects.

TweetNER7 #1969

Omiss.

“Hi ! I ’m originally an iKONIC and a teume since ygtb era and it ’s just sad that i do n’t any teume / iKONIC moots It ’s my first stan account so please help me gain some moots. Pls rt & ff me,I will definitely ff you back {{USERNAME}}”

Gold	T-NER	GLiNER2	GPT-5.4-mini
iKONIC/per	iKONIC/grp	USERNAME/per iKONIC/prod moots/prod stan account/prod teume/prod	iKONIC/grp teume/grp ygtb/cw
	F_1 .000 $\hat{F}_1$.612 U .000	F_1 .000 $\hat{F}_1$.546 U .600	F_1 .000 $\hat{F}_1$.530 U .667

Gold labels “iKONIC” as a person, but it is a K-pop fan group name; models labeling it as “group” are arguably more correct than the gold annotation.

TweetNER7 #2195

Omiss.

“Random It is predicted that the reason why night insects,such as moths,are attracted to lights is because they mistake them for the light of the moon,which they used to navigate the Earth before mankind made artificial lights. { @MISSIO@ } # CanYouFeelTheSun {{USERNAME}} # alt18”

Gold	T-NER	GLiNER2	GPT-5.4-mini
{ @MISSIO@ }/grp	{ @MISSIO@ }/per	Earth/loc artificial lights/prod lights/prod mankind/grp	CanYouFeelTheSun/cw Earth/loc
	F_1 .000 $\hat{F}_1$.612 U .000	F_1 .000 $\hat{F}_1$.000 U 1.00	F_1 .000 $\hat{F}_1$.000 U 1.00

TweetNER7 #2250

Alias · Omiss.

“HONESTLY I’M VERY CONCERNED PLEASE VOTE FOR TSA !!! THIS IS ONE WE ABSOLUTELY CANNOT LOSE # BTS { @BTS_twt@ } {{URL}}”

Gold	T-NER	GLiNER2	GPT-5.4-mini
TSA/grp { @BTS_twt@ }/grp	TSA/grp { @BTS_twt@ }/grp	TSA/corp	BTS/grp TSA/grp
	F_1 1.00 $\hat{F}_1$ 1.00 U .000	F_1 .000 $\hat{F}_1$.471 U .000	F_1 .500 $\hat{F}_1$.616 U .000

GPT predicts “BTS” for the gold handle “{ @BTS_twt@ }”:the same referent under a different surface form.

TweetNER7 #2428

Alias · Omiss.

“ ‘ THE OCEAN EYES ’ I love the sound of this. { @mewsuppasit@ } # MewSuppasit #[Thai]”

Gold	T-NER	GLiNER2	GPT-5.4-mini
MewSuppasit/grp THE OCEAN EYES/cw { @mewsuppasit@ }/per	MewSuppasit/per THE OCEAN EYES/cw { @mewsuppasit@ }/per	THE OCEAN EYES/cw	MewSuppasit/per THE OCEAN EYES/cw [Thai]/per
	F_1 .667 $\hat{F}_1$.885 U .000	F_1 .500 $\hat{F}_1$.500 U .000	F_1 .333 $\hat{F}_1$.779 U .333

GPT predicts the Thai name [Thai] of the same person as the gold handle alias. Gold also labels “MewSuppasit” as a group rather than a person.

TweetNER7 #2547

Alias · Omiss.

“I hope you live in a place where the sky is blue enough,the sunshine is good enough,the shops on the street are clean and delicious,and the lights are always on at home. { @mewsuppasit@ } # MewSuppasit #[Thai]”

Gold	T-NER	GLiNER2	GPT-5.4-mini
{ @mewsuppasit@ }/per	MewSuppasit/per { @mewsuppasit@ }/per	home/loc lights/prod shops/loc sky/loc street/loc sunshine/loc	MewSuppasit/per mewsuppasit/per [Thai]/per
	F_1 .667 $\hat{F}_1$.946 U .000	F_1 .000 $\hat{F}_1$.000 U 1.00	F_1 .000 $\hat{F}_1$.812 U .333

Same pattern as example TweetNER7#2428 above

TweetNER7 #2631

Alias · Partial

“The developer and landlord class will not cease until Dublin City is nothing more than a series of hotels An Bord Pleanála : Save James Joyce ’s ’ House of the Dead ’ (Usher ’s Island) from becoming a tourist hostel. - Sign the Petition ! {{URL}} via {{@Change org [Cyrillic]@}}”

Gold	T-NER	GLiNER2	GPT-5.4-mini
Dublin City/loc House of the Dead/cw Island/loc James Joyce/per Usher/loc developer and landlord class/grp {@Change org [Cyrillic]@}/prod	An Bord Pleanála/grp Dublin City/loc House of the Dead/cw James Joyce/per Usher ’s Island/loc {@Change org [Cyrillic]@}/prod	Dublin City/loc House of the Dead/cw James Joyce/per developer/corp	An Bord Pleanála/corp Change org Russia/corp Dublin City/loc House of the Dead/cw James Joyce/per Usher’s Island/loc
F_1 .615 $\hat{F}_1$.838 U .167 F_1 .545 $\hat{F}_1$.571 U .250 F_1 .462 $\hat{F}_1$.719 U .333			

GPT predicts “Change org Russia.” which is an alias for “{@Change org [Cyrillic]@}”

TweetNER7 #2711

Alias · Partial

“Work hard,have fun,make history {{USERNAME}} WE LOVE YOU SOHEL #HBDIsmartSohel #HBDSyedSohelRyan”

Gold	T-NER	GLiNER2	GPT-5.4-mini
HBDIsmartSohel/per HBDSyedSohelRyan/per	HBDIsmartSohel/grp HBDSyedSohelRyan/grp SOHEL/per	SOHEL/per	Sohel/per Syed Sohél Ryan/per
F_1 .000 $\hat{F}_1$.716 U .000 F_1 .000 $\hat{F}_1$.775 U .000 F_1 .000 $\hat{F}_1$.847 U .000			

GPT extracts “Sohel” and “Syed Sohél Ryan” from birthday hashtags: correct referents that strict matching rejects because the surface forms differ from the hashtag gold.

TweetNER7 #2776

Alias · Omiss.

“Flawless performance by Izzy {@Israel Adesanya@}. {@Paulo Costa (Borrachinha)@} looked defeated before the fight begun. To his corner - There ’s no point in hyping your fighter hoping he wins and you get a piece of that cake. Confidence comes with experience .”

Gold	T-NER	GLiNER2	GPT-5.4-mini
{@Israel Adesanya@}/per {@Paulo Costa (Borrachinha)@}/per	fight/evt {@Israel Adesanya@}/per {@Paulo Costa (Borrachinha)@}/per	Borrachinha/per Izzy/per cake/prod corner/loc fight/evt fighter/per	Borrachinha/per Israel Adesanya/per Izzy/per Paulo Costa/per
F_1 .800 $\hat{F}_1$.874 U .333 F_1 .000 $\hat{F}_1$.622 U .667 F_1 .000 $\hat{F}_1$.826 U .000			

“Izzy” is a nickname for gold “Israel Adesanya,” and “Borrachinha”/“Paulo Costa” are alternate forms of the compound gold name. Only semantic scoring credits these valid aliases.

TweetNER7 #2793

Alias · Type

“I have spent a dumb amount of time mapping out my first water cooling loop. I keep modifying it every few days. Really need to make a decision before {@Cyberpunk 2077@} comes out .”

Gold	T-NER	GLiNER2	GPT-5.4-mini
{@Cyberpunk 2077@}/prod	{@Cyberpunk 2077@}/cw	water cooling loop/prod	Cyberpunk 2077/cw
F_1 .000 $\hat{F}_1$.714 U .000 F_1 .000 $\hat{F}_1$.000 U 1.00 F_1 .000 $\hat{F}_1$.650 U .000			

SciERC

SciERC #47

Type · Partial · Omiss.

“Towards deep analysis of compositional classes of paraphrases, we have examined a class-oriented framework for collecting paraphrase examples, in which sentential paraphrases are collected for each paraphrase class separately by means of automatic candidate generation and manual judgement. Our preliminary experiments on building a paraphrase corpus have so far been producing promising results, which we have evaluated according to cost-efficiency, exhaustiveness, and reliability.”

Gold	T-NER	GLiNER2	GPT-5.4-mini
automatic candidate generation/ <i>Meth</i> class-oriented framework/ <i>Meth</i> compositional classes of paraphrases/ <i>Task</i> manual judgement/ <i>Meth</i> paraphrase corpus/ <i>Mat</i> paraphrase examples/ <i>Mat</i> sentential paraphrases/ <i>Mat</i>	automatic candidate generation/ <i>Meth</i> class-oriented framework/ <i>Meth</i> compositional classes of paraphrases/ <i>OST</i> deep analysis of/ <i>Task</i> manual judgement/ <i>OST</i> paraphrase/ <i>OST</i> paraphrase corpus/ <i>OST</i> paraphrase examples/ <i>OST</i> sentential paraphrases/ <i>OST</i>	automatic candidate generation/ <i>Meth</i> cost-efficiency/ <i>Met</i> exhaustiveness/ <i>Met</i> experiments/ <i>Task</i> manual judgement/ <i>Meth</i> paraphrase corpus/ <i>Gen</i> paraphrases/ <i>Gen</i> reliability/ <i>Met</i> sentential paraphrases/ <i>OST</i>	automatic candidate generation/ <i>Meth</i> class-oriented framework/ <i>Meth</i> compositional classes of paraphrases/ <i>OST</i> cost-efficiency/ <i>Met</i> deep analysis/ <i>Task</i> exhaustiveness/ <i>Met</i> manual judgement/ <i>Meth</i> paraphrase class/ <i>OST</i> paraphrase corpus/ <i>OST</i> paraphrase examples/ <i>OST</i> reliability/ <i>Met</i> sentential paraphrases/ <i>OST</i>
	F_1 .250 $\hat{F}_1$.477 U .778	F_1 .250 $\hat{F}_1$.382 U .778	F_1 .316 $\hat{F}_1$.500 U .750

All models struggle with SciERC’s fine-grained type distinctions (Task versus OtherScientificTerm).

SciERC #56

Type · Partial · Omiss.

“We investigate the problem of learning to predict moves in the board game of Go from game records of expert players. In particular, we obtain a probability distribution over legal moves for professional play in a given position. This distribution has numerous applications in computer Go, including serving as an efficient stand-alone Go player. It would also be effective as a move selector and move sorter for game tree search and as a training tool for Go players. Our method has two major components : a) a pattern extraction scheme for efficiently harvesting patterns of given size and shape from expert game records and b) a Bayesian learning algorithm (in two variants) that learns a distribution over the values of a move given a board position based on the local pattern context. The system is trained on 181,000 expert games and shows excellent prediction performance as indicated by its ability to perfectly predict the moves made by professional Go players in 34 % of test positions .”

Gold	T-NER	GLiNER2	GPT-5.4-mini
Bayesian learning algorithm/ <i>Meth</i> Go players/ <i>OST</i> It/ <i>Gen</i> board game of Go/ <i>Task</i> computer Go/ <i>OST</i> distribution/ <i>Gen</i> expert games/ <i>Mat</i> game records of expert players/ <i>Mat</i> game tree search/ <i>Meth</i> local pattern context/ <i>OST</i> method/ <i>Gen</i> move selector/ <i>Meth</i> move sorter/ <i>Meth</i> pattern extraction scheme/ <i>Meth</i> predict/ <i>Task</i> probability distribution/ <i>OST</i> stand-alone/ <i>Meth</i> training/ <i>Task</i>	Bayesian learning algorithm/ <i>Meth</i> Go/ <i>Meth</i> It/ <i>Gen</i> board game/ <i>Task</i> computer Go/ <i>Task</i> distribution/ <i>Gen</i> game tree search/ <i>Task</i> method/ <i>Gen</i> move selector/ <i>Meth</i> move sorter/ <i>Meth</i> pattern extraction scheme/ <i>Meth</i> predict/ <i>Task</i> probability distribution/ <i>OST</i> stand-alone/ <i>Meth</i> training/ <i>Task</i>	Bayesian learning algorithm/ <i>Meth</i> Go/ <i>Gen</i> LRB/ <i>Meth</i> RRB/ <i>Meth</i> computer Go/ <i>Gen</i> expert players/ <i>Gen</i> game records/ <i>Mat</i> game tree search/ <i>Task</i> legal moves/ <i>OST</i> local pattern context/ <i>OST</i> move selector/ <i>Task</i> move sorter/ <i>Task</i> pattern extraction scheme/ <i>Meth</i> prediction performance/ <i>Met</i> probability distribution/ <i>OST</i> professional play/ <i>Task</i> training tool/ <i>Task</i>	181,000 expert games/ <i>Met</i> 34 %/ <i>Met</i> Bayesian learning algorithm/ <i>Meth</i> Go/ <i>Gen</i> Go player/ <i>Gen</i> board position/ <i>Gen</i> computer Go/ <i>Gen</i> distribution/ <i>OST</i> excellent prediction performance/ <i>Met</i> expert players/ <i>Gen</i> game records/ <i>OST</i> game tree search/ <i>Task</i> legal moves/ <i>Task</i> local pattern context/ <i>OST</i> move selector/ <i>Meth</i> move sorter/ <i>Meth</i> pattern extraction scheme/ <i>Meth</i> patterns/ <i>OST</i> test positions/ <i>Met</i> training tool/ <i>Meth</i>
	F_1 .485 $\hat{F}_1$.693 U .133	F_1 .286 $\hat{F}_1$.691 U .353	F_1 .263 $\hat{F}_1$.648 U .350

Dense entity extraction with many partial overlaps (“Go” versus “Go players,” “game records” versus “game records of expert

players”). Soft scoring captures proportional credit that strict matching denies.

SciERC #64

Alias · Type · Partial · Omiss.

“This work proposes a new research direction to address the lack of structures in traditional n -gram models. It is based on a weakly supervised dependency parser that can model speech syntax without relying on any annotated training corpus. Labeled data is replaced by a few hand-crafted rules that encode basic syntactic knowledge. Bayesian inference then samples the rules, disambiguating and combining them to create complex tree structures that maximize a discriminative model’s posterior on a target unlabeled corpus. This posterior encodes sparse se-lectional preferences between a head word and its dependents. The model is evaluated on English and Czech newspaper texts, and is then validated on French broadcast news transcriptions.”

Gold	T-NER	GLiNER2	GPT-5.4-mini
Bayesian inference/Meth English and Czech newspaper texts/Mat French broadcast news transcriptions/Mat Labeled data/OST annotated training corpus/Mat complex tree structures/OST discriminative model’s posterior/OST hand-crafted rules/OST lack of structures/OST lack of structures in traditional n -gram models/Task model/Gen n -gram models/Meth posterior/Gen rules/OST sparse se-lectional preferences/OST speech syntax/OST syntactic knowledge/OST them/Gen unlabeled corpus/Mat weakly supervised dependency parser/Task	Bayesian inference/Meth English and Czech newspaper texts/Mat It/Gen Labeled data/OST annotated training corpus/Mat complex tree structures/OST discriminative model/Meth hand-crafted rules/OST head/OST model/Gen n -gram models/Meth posterior/OST sparse se-lectional preferences/OST speech syntax/OST syntactic knowledge/OST unlabeled corpus/Mat weakly supervised dependency parser/Meth	Bayesian inference/Meth Czech/Gen English/Gen dependency parser/Meth dependents/OST head word/OST n -gram models/OST posterior/Met se-lectional preferences/OST speech syntax/OST syntactic knowledge/OST training corpus/Mat tree structures/OST	Bayesian inference/Meth Czech/Gen English/Gen French/Gen annotated training corpus/Gen broadcast news transcriptions/Gen dependents/Gen discriminative model/Meth hand-crafted rules/Meth head word/OST n -gram models/OST newspaper texts/Gen posterior/Met rules/Gen selectional preferences/OST speech syntax/OST syntactic knowledge/OST target unlabeled corpus/Gen tree structures/OST weakly supervised dependency parser/Meth
	F_1 .649 $\hat{F}_1$.839 U .000	F_1 .182 $\hat{F}_1$.577 U .154	F_1 .150 $\hat{F}_1$.581 U .150

SciERC #69

Type

“The goal of this work is the enrichment of human-machine interactions in a natural language environment. Because a speaker and listener can not be assured to have the same beliefs, contexts, perceptions, backgrounds, or goals, at each point in a conversation, difficulties and mistakes arise when a listener interprets a speaker’s utterance. These mistakes can lead to various kinds of misunderstandings between speaker and listener, including reference failures or failure to understand the speaker’s intention. We call these misunderstandings miscommunication. Such mistakes can slow, and possibly break down, communication. Our goal is to recognize and isolate such miscommunications and circumvent them. This paper highlights a particular class of miscommunication – reference problems – by describing a case study and techniques for avoiding failures of reference. We want to illustrate a framework less restrictive than earlier ones by allowing a speaker leeway in forming an utterance about a task and in determining the conversational vehicle to deliver it. The paper also promotes a new view for extensional reference.”

Gold	T-NER	GLiNER2	GPT-5.4-mini
extensional	human-machine/Task	backgrounds/OST	case study/Meth
reference/Task	miscommunication/OST	beliefs/OST	extensional
failures of	miscommunications/OST	case study/Meth	reference/OST
reference/Task	misunderstandings/OST	contexts/OST	human-machine
human-machine	natural language/OST	extensional	interactions/Task
interactions/Task	reference/OST	reference/OST	miscommunication/OST
miscommunication/Task		framework/OST	natural language
miscommunications/Task		goals/OST	environment/OST
natural language		human-machine	reference problems/Task
environment/OST		interactions/OST	
reference failures/OST		miscommunication/OST	
reference problems/Task		misunderstandings/OST	
speaker’s		perceptions/OST	
intention/OST		reference failures/OST	
techniques/Gen		reference problems/OST	
them/Gen		task/Task	
		techniques/Meth	
		utterance/Task	
F_1 .000 $\hat{F}_1$.403 U .000 F_1 .074 $\hat{F}_1$.571 U .250 F_1 .353 $\hat{F}_1$.608 U .167			

T-NER scores zero strict F_1 despite predicting most gold entities, solely because type mismatches (Task versus OtherScientificTerm). Soft scoring recovers to ≈ 0.4 . GPT and GLiNER2 retrieve better results.

SciERC #66

Alias · Type · Partial

“We present a novel method for discovering parallel sentences in comparable,non-parallel corpora. We train a maximum entropy classifier that,given a pair of sentences,can reliably determine whether or not they are translations of each other. Using this approach,we extract parallel data from large Chinese,Arabic,and English non-parallel newspaper corpora. We evaluate the quality of the extracted data by showing that it improves the performance of a state-of-the-art statistical machine translation system. We also show that a good-quality MT system can be built from scratch by starting with a very small parallel corpus (100,000 words) and exploiting a large non-parallel corpus. Thus,our method can be applied with great benefit to language pairs for which only scarce resources are available .”

Gold	T-NER	GLiNER2	GPT-5.4-mini
Chinese,Arabic,and English non-parallel newspaper corpora/Mat MT system/Meth approach/Gen comparable,non-parallel corpora/Mat discovering parallel sentences/Task it/Gen maximum entropy classifier/Meth method/Gen non-parallel corpus/Mat parallel corpus/Mat parallel data/Mat scarce resources/Mat statistical machine translation system/Meth	Chinese,Arabic,and English non-parallel newspaper corpora/Mat MT system/Meth approach/Gen comparable,non-parallel corpora/Mat it/Gen maximum entropy classifier/Meth method/Gen parallel sentences/Mat statistical machine translation system/Meth	Arabic/Gen Chinese/Gen English/Gen MT system/Meth language pairs/Gen maximum entropy classifier/Meth performance/Met statistical machine translation system/Meth translations/Task	100,000 words/Met Arabic/Gen Chinese/Gen English/Gen MT system/Task comparable/OST language pairs/Gen maximum entropy classifier/Meth method/Meth non-parallel corpora/Task non-parallel corpus/Task non-parallel newspaper corpora/Task pair of sentences/Gen parallel corpus/Task parallel data/Task parallel sentences/Task quality/Met scarce resources/OST state-of-the-art statistical machine translation system/Task this approach/Meth translations/OST
	F_1 .727 $\hat{F}_1$.809 U .000	F_1 .273 $\hat{F}_1$.441 U .222	F_1 .059 $\hat{F}_1$.547 U .238

SciERC #85

Alias · Type · Paraph. · Omiss.

“Sentence boundary detection in speech is important for enriching speech recognition output, making it easier for humans to read and downstream modules to process. In previous work, we have developed hidden Markov model (HMM) and maximum entropy (Maxent) classifiers that integrate textual and prosodic knowledge sources for detecting sentence boundaries. In this paper, we evaluate the use of a conditional random field (CRF) for this task and relate results with this model to our prior work. We evaluate across two corpora (conversational telephone speech and broadcast news speech) on both human transcriptions and speech recognition output. In general, our CRF model yields a lower error rate than the HMM and Max-ent models on the NIST sentence boundary detection task in speech, although it is interesting to note that the best results are achieved by three-way voting among the classifiers. This probably occurs because each model has different strengths and weaknesses for modeling the knowledge sources.”

Gold	T-NER	GLiNER2	GPT-5.4-mini
CRF model/ <i>Meth</i> HMM and Max-ent models/ <i>Meth</i> NIST sentence boundary detection task/ <i>Mat</i> Sentence boundary detection/ <i>Task</i> broadcast news speech/ <i>Mat</i> classifiers/ <i>Meth</i> conditional random field (CRF)/ <i>Meth</i> conversational telephone speech/ <i>Mat</i> corpora/ <i>Gen</i> detecting sentence boundaries/ <i>Task</i> error rate/ <i>Met</i> hidden Markov model (HMM) and maximum entropy (Maxent) classifiers/ <i>Meth</i> human transcriptions/ <i>OST</i> it/ <i>Gen</i> knowledge sources/ <i>Mat</i> model/ <i>Gen</i> speech/ <i>Mat</i> speech recognition output/ <i>OST</i> task/ <i>Gen</i> textual and prosodic knowledge sources/ <i>Mat</i> three-way voting/ <i>Meth</i>	Sentence boundary detection in/ <i>Task</i> conditional random field (CRF)/ <i>Meth</i> conversational telephone speech/ <i>Mat</i> detecting/ <i>Task</i> hidden Markov model (HMM)/ <i>Meth</i> maximum entropy (Maxent) classifiers/ <i>Meth</i> model/ <i>Gen</i> sentence boundaries/ <i>OST</i> speech/ <i>Mat</i> speech recognition output/ <i>Task</i> task/ <i>Gen</i> textual and prosodic knowledge sources/ <i>OST</i>	Sentence boundary detection/ <i>Task</i> conditional random field/ <i>Meth</i> corpora/ <i>Mat</i> error rate/ <i>Met</i> human transcriptions/ <i>Mat</i> humans/ <i>Gen</i> knowledge sources/ <i>OST</i> speech/ <i>Gen</i> three-way voting/ <i>Meth</i>	CRF/ <i>Meth</i> HMM/ <i>Meth</i> Max-ent/ <i>Meth</i> Maxent/ <i>Meth</i> NIST sentence boundary detection task in speech/ <i>Task</i> Sentence boundary detection in speech/ <i>Task</i> broadcast news speech/ <i>Gen</i> classifiers/ <i>Meth</i> conditional random field/ <i>Meth</i> conversational telephone speech/ <i>Gen</i> error rate/ <i>Met</i> hidden Markov model/ <i>Meth</i> human transcriptions/ <i>Gen</i> knowledge sources/ <i>OST</i> maximum entropy/ <i>Meth</i> speech recognition output/ <i>Task</i> three-way voting/ <i>Meth</i>
	F_1 .303 $\hat{F}_1$.708 U .083	F_1 .200 $\hat{F}_1$.705 U .000	F_1 .158 $\hat{F}_1$.595 U .059

All models extract abbreviation aliases (“CRF” for “CRF model,” “HMM” for the full name) that strict matching rejects. Soft scoring recovers substantial credit across all three.